

Fat-Soluble Vitamins



Vitamin	Function	Sources
Vitamin E	Similar to vitamin A, vitamin E is a class of compounds. The most common ones are known as the tocopherols and the tocotrienols, each of which have four types: alpha, beta, gamma, and delta. Vitamin E is one of the main antioxidant molecules in the body and protects cells from oxidative damage. There is no robust evidence indicating that vitamin E can improved athletic performance.	• Almonds • Sunflower seeds • Wheat germ • Hazelnuts • Pine nuts • Salmon • Avocado • Trout
Vitamin A	Vitamin A is a group of compounds referred to as retinoids. The most common retinoids are retinol, retinal, and retinoic acid. Furthermore, vitamin A can be made from beta-carotene and other similar compounds known as the carotenoids which are antioxidants. Vitamin A plays several critical roles in human health. It is critical in vision, immune function, and for human development. There does not appear to be any evidence to suggest that vitamin A supplementation improves athletic performance.	• Liver • Cod liver oil • Mackerel • Salmon • Tuna • Butter • Goat cheese • Eggs • Sweet potato • Carrots • Squash • Kale
Vitamin D	Vitamin D plays many critical roles in human health, including immune function, bone formation, and calcium metabolism. Vitamin D insufficiency was observed in roughly 56% of young athletes, which risks being elevated at higher latitudes where there is less sun exposure. These low levels of vitamin D may impair some aspects of performance and recovery. There is some evidence that correcting these deficiencies can improve aerobic performance, muscle strength, and recovery from training. However, much more work needs to be done to determine how much benefit athletes may get from vitamin D supplementation.	• Salmon • Sardines • Mackerel • Tuna • Cod liver oil • Shiitake mushroom • Fortified milk • Fortified cheese

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Vitamin K	Vitamin K is the name for a family of compounds that have a similar chemical structure. The compounds in this family are known as the phyloquinones (vitamin K1) and menaquinones (vitamin K2). Broadly speaking vitamin K is involved in regulating blood homeostasis by controlling clotting. It also plays critical roles in bone formation and remodeling as well as working synergistically with vitamin D. Currently, there is no evidence to suggest that vitamin K supplementation, per se, may benefit athletic performance.	• Kale • Collard greens • Mustard greens • Swiss chard • Mustard parsley • Romaine • Green leaf lettuce • Brussel sprouts